

Tasks (100 marks)

- Local inference **40 marks**
 - Collect user input (10)
 - Infer locally and display result (15)
 - Run on emulated/physical IoT device (15)
- Cloud inference **30 marks**
 - Run inference in cloud virtual machine (10)
 - Communicate btw IoT device & cloud (20)
- Advanced tasks **30 marks**
 - Train your own model (10)
 - Support multiple concurrent users (10)
 - Support online model updating (10)

Marking Criteria (1)

- Collect user input (10)
 - **IF** load input from storage (5)
 - **ELSE** collect real-time input by touch screen, microphone, camera, built-in sensors (10)

Marking Criteria (2)

- Infer locally and display result (15)
 - **IF** your app offloads the execution to cloud, (10)
 - **ELSE** run heuristic algorithm (not neural network) on the mobile app (5)
 - Display the inference result in real time by screen or synthetic voice (5)

Marking Criteria (3)

- Run on emulated/physical IoT device (15)
 - **IF** the program runs natively on a desktop or laptop computer (5)
 - **ELSE** the program runs on an emulated or a physical IoT device (15)
 - iOS emulator provided by Xcode
 - Android emulator
 - Real smartphone
 - Raspberry Pi + add-on sensors (camera, microphone, etc)

Marking Criteria (4)

- Run inference in cloud virtual machine (10)
 - **IF** deploy server program on a cloud virtual machine, e.g., Azure. (10)
 - **ELSE** deploy server program on your own computer (10)

Marking Criteria (5)

- Communication btw IoT device and cloud (20)
 - **IF** HTML-based communications (5)
 - **ELSE**
 - Send the user input from the mobile app to the cloud for inference (10)
 - Send the inference result from the cloud to the mobile app (10)

Marking Criteria (6)

- Train your own model (10)
 - **IF** train the used ML model by yourself (10)
 - Training program should be in the code package
 - Training results (e.g., accuracy) in a readme file
 - The training data can be any publicly available dataset
 - **ELIF** use downloaded pre-trained model (5)
 - **ELSE** use heuristic algorithm (not neural network) (2)

Marking Criteria (7)

- Support multiple concurrent users (10)
 - Demonstrate multiple IoT devices can use the cloud service simultaneously

Marking Criteria (8)

- Online model updating (10)
 - Demonstrate that the model in the cloud can be updated at run time using newly collected user data
 - User input and the corresponding label are transmitted to cloud
 - The cloud retrains the model